Modelling Assumptions

Exam code: 9MA0

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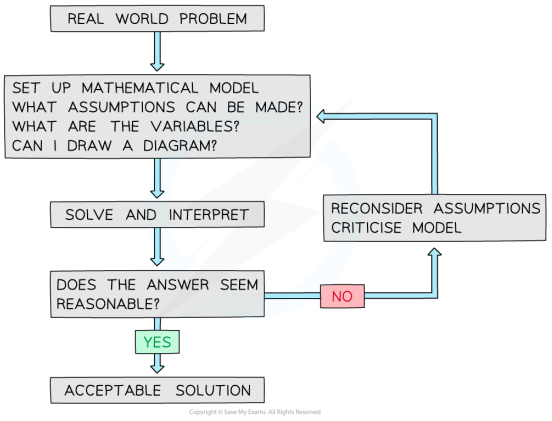
Modelling assumptions

What is modelling in Mechanics?

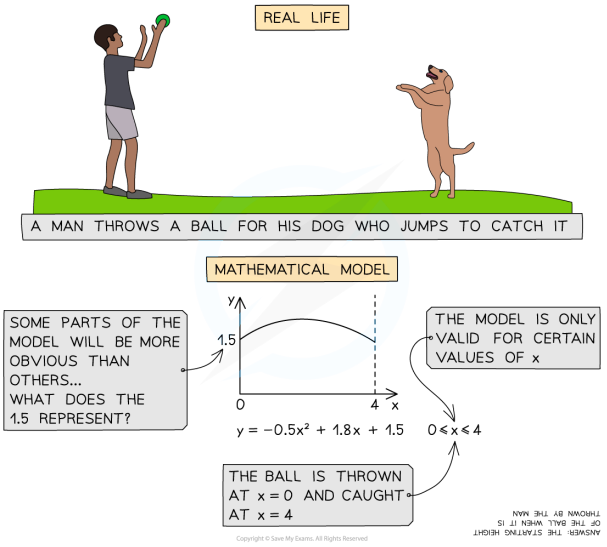
- Mechanics uses **modelling** to solve problems in real life situations
- We use **assumptions** to simplify real life problems in order to turn them into equations or graphs that can be solved
- We will sometimes need to **criticise** or **refine our assumptions** to improve the model



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

What is a modelling assumption?

- The assumptions you make will affect the calculations in each problem differently



What types of modelling assumptions do I need to know?

- **Gravity** is constant and acts vertically downwards
- **Air resistance** is usually modelled as **negligible** and can be ignored
- A **smooth** surface has no **friction**
- A **rough** surface has a **frictional force** between the surface and any object that makes contact with it
- A **particle** has **negligible** dimensions, therefore forces will all act on a particle at the same point
- A **rod** or a **beam** should be treated as a long, rigid particle
- A **uniform** object's **mass** is distributed evenly
- A **light** object has zero **mass**
- An **inextensible** object cannot be stretched

Worked Example

Two blocks, A and B are attached by means of a light inextensible string running over a smooth pulley. Block A has mass 0.7 kg and is accelerating along a smooth horizontal surface, block B has a mass 0.2 kg. Both A and B are modelled as particles. State how the following modelling assumptions can be used in your calculations.

a) A and B are both particles.

b) The string is light.

c) The string is inextensible.

d) The pulley is smooth.

e) The surface A is moving along is smooth.

Particles have no dimensions so air resistance cannot act on the object

b) The tension is constant throughout the string (same at A and B)

Light objects have zero mass

c) Acceleration will be the same for both blocks

Inextensible means the string cannot stretch

d) The tension in the string is the same on both sides of the pulley

Smooth surfaces provide no friction

e) There is no frictional force between the surface and block A

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Make sure you fully understand the definitions of all the words in this section so that you can be clear about what your exam question is asking of you.

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